Number Bonds to 10 and 20

Find the missing numbers. Use the ten frames and part-whole models to help.

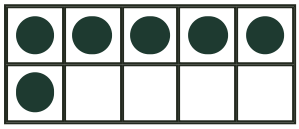
Name _____ Date _____ Class _____

Learning objective: I can recall and use number bonds to 10 and 20.

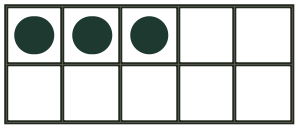
✓ I know pairs of numbers that make 10 ✓ I can use bonds to 10 to find bonds to 20 ✓ I can find a missing number

1 Ten frames

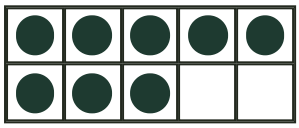
Count the dots. How many more make 10? Write the number sentence.



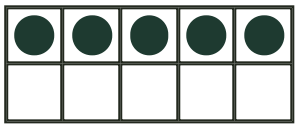
☐ + ☐ = 10



☐ + ☐ = 10



☐ + ☐ = 10



☐ + ☐ = 10

2 Bonds to 10

Write the missing number in each box.

1. 3 + ☐ = 10
2. ☐ + 6 = 10
3. 8 + ☐ = 10
4. 10 = 5 + ☐
5. ☐ + 1 = 10
6. 10 − 4 = ☐
7. 10 − ☐ = 3
8. 2 + ☐ = 10
9. 10 = ☐ + 9
10. 0 + ☐ = 10
11. 10 − 7 = ☐
12. ☐ + 4 = 10

3 All the bonds to 10

Complete the pattern. What do you notice?

0 + ☐ = 10 1 + ☐ = 10 2 + ☐ = 10 3 + ☐ = 10

4 + ☐ = 10 5 + ☐ = 10 6 + ☐ = 10 7 + ☐ = 10

8 + ☐ = 10 9 + ☐ = 10 10 + ☐ = 10

Challenge: how many different pairs of numbers make 10? (3 + 7 and 7 + 3 count as the same pair.) ☐

Number Bonds to 20

4 Use bonds to 10 If you know a bond to 10, you know a bond to 20!

3 + 7 = 10

so…

13 + 7 = ☐

6 + 4 = 10

so…

16 + ☐ = 20

2 + 8 = 10

so…

☐ + 8 = 20

5 Bonds to 20 Write the missing number in each box.

1 14 + ☐ = 20

2 ☐ + 11 = 20

3 20 − 5 = ☐

4 20 = 13 + ☐

5 18 + ☐ = 20

6 ☐ + 10 = 20

7 20 − ☐ = 8

8 9 + ☐ = 20

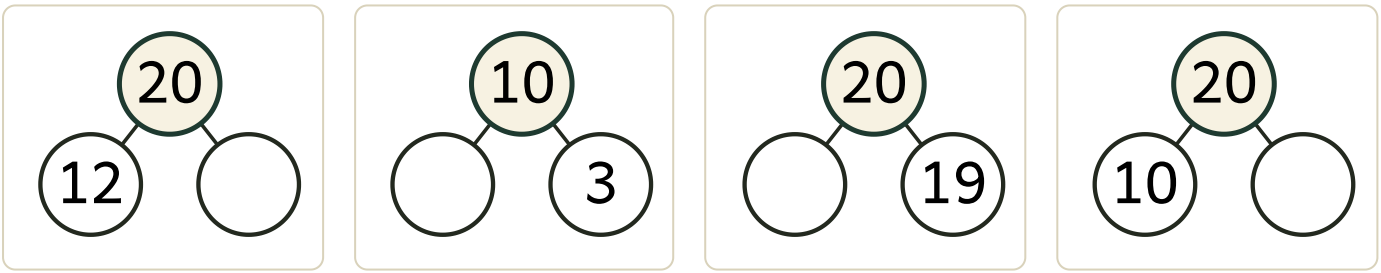
9 20 − 17 = ☐

10 ☐ + 15 = 20

11 20 = ☐ + 6

12 20 − 0 = ☐

6 Part-whole models The top circle is the whole. Write the missing part.



7 Reasoning Sam says, “11 + 8 = 20.” Is Sam right? Explain how you know.

Answer Key

Number Bonds to 10 and 20

Question	Answers
1 Ten frames	6 + 4 = 10; 3 + 7 = 10; 8 + 2 = 10; 5 + 5 = 10
2 Bonds to 10	(1) 7; (2) 4; (3) 2; (4) 5; (5) 9; (6) 6; (7) 7; (8) 8; (9) 1; (10) 10; (11) 3; (12) 6
3 All the bonds to 10	0+10, 1+9, 2+8, 3+7, 4+6, 5+5, 6+4, 7+3, 8+2, 9+1, 10+0. Pattern: as the first number goes up by 1, the second number goes down by 1. **Challenge:** 6 different pairs – 0+10, 1+9, 2+8, 3+7, 4+6, 5+5.
4 Use bonds to 10	13 + 7 = **20**; 16 + **4** = 20; **12** + 8 = 20
5 Bonds to 20	(1) 6; (2) 9; (3) 15; (4) 7; (5) 2; (6) 10; (7) 12; (8) 11; (9) 3; (10) 5; (11) 14; (12) 20
6 Part-whole models	20 = 12 + **8**; 10 = **7** + 3; 20 = **1** + 19; 20 = 10 + **10**
7 Reasoning	No. 11 + 8 = 19, not 20. Possible explanations: 1 + 8 = 9, not 10, so 11 + 8 = 19; or 11 + 9 = 20 / 12 + 8 = 20.

Teaching note: encourage children to say each fact both ways (3 + 7 and 7 + 3) and to spot the link between 3 + 7 = 10 and 13 + 7 = 20. Sections 1–3 suit Year 1; sections 4–7 extend to Year 2 (recall and use addition and subtraction facts to 20).